Real-Time Moving Vehicle Detection in Satellite Video Based on Historical Differential Information and Grouping Features

Cong Xu, Hua Qi, Yahui Zheng, and Shiyong Peng

***Abstract*— Due to the characteristics of small size, few color texture features, and blurred object boundaries of moving vehicles in satellite video, traditional methods based on background subtraction or frame difference and deep learning methods based on single-frame images continue to have a high number of false positives (FPs) and false negatives (FNs). In this article, a satellite video moving vehicle detection network, consisting of the historical frame differential module (HFDM) and the improved YOLOv5 algorithm, is proposed. Initially, a multiscene mosaic (MSM) data augmentation strategy is employed to increase the data complexity, thereby improving the detection ability of the model in complex scenes. Then, HFDM is used to extract the motion information in the historical frames and fuse the motion features with the current frame to obtain the information on the spatio-temporal feature, which can compensate for the absence of single-frame image features. Finally, grouping features of multiple consecutive frames are calculated independently using grouping convolution, the down-sampling rate of the network model is reduced, large-scale feature maps are used for prediction, group attention (GA) and path aggregation network (PANet) are used instead to improve the grouping feature expression of consecutive frames, named GA-PANet. Experiments indicate that the method in this article achieves better detection performance and meets the real-time requirements on both test datasets I and II. The method can deal with the problems of false alarms caused by weak color texture features and blurry boundaries and achieve state-of-the-art performance in real-time moving vehicle detection in satellite video.**

***Index Terms*— Historical frame differential module (HFDM), moving vehicle, multiscene mosaic (MSM) data augmentation, real-time object detection, satellite video, YOLOv5.**

I. Introduction

VIDEO satellite is a type of Earth observation satellite that has emerged in recent years, capturing video sequences of particular regions and having the capacity to conduct continuous surveillance of particular scenes. Video satellite has significant applications in urban planning, traffic monitoring, and military reconnaissance, continuous monitoring of global hotspot areas and objects utilizing satellites to obtain dynamic information has become essential, and moving object detection in satellite video has become a central issue in the development of remote sensing satellites [45], [46]. By detecting objects of interest in real-time, it contributes to urban management, scheduling, and monitoring, and improves urban planning and security alerting capabilities.

Skysat and Jilin-1 are the two most commonly used satellite video data by researchers at present. In 2013 and 2014, Skybox launched SkySat-1 and SkySat-2, which can capture full-color video with 1.1 m of spatial resolution for up to 90 s [2]. The Jilin-1 series of satellites, developed by Chang Guang Satellite Technology Company Ltd. (CGSTL), which could shoot full-color video with a resolution of nearly 1 m, the video frame rate of the Jilin1-1 and Jilin1-2 is 25 frames/s (FPS), the video frame rate of the Jilin1-3 is 10 or 15 FPS [3]. The characteristics of satellite video captured content are as follows [4], [5], [6]. 1) In the captured satellite video, the background of adjacent frames, such as buildings and roads, hardly shifts due to the high gaze effect. 2) Moving vehicles are characterized by their relatively small size, typically occupying only a few or a dozen pixels, and their limited color texture features. 3) It is difficult to distinguish the moving vehicle boundaries because the captured video has blurring, complex background, and low contrast between vehicle and background. Fig. 1 shows an example of the VISO [36] Jilin-1 satellite video dataset, which shows that detecting tiny moving vehicles on an extensive range of satellite video is challenging.

Traditional methods for video object detection include background subtraction [7], [18], [43], [44], frame difference [5], [6], and the optical flow [8], [9], [10]. The background subtraction [7] approach models the background utilizing multiple frames and then calculates the difference between the current frame image and the background image to generate the moving objects. The frame difference [5] is accomplished mostly by calculating differential features and thresholding to extract motion regions. The optical flow calculates the motion field of the pixel point and analyzes the time-domain variation and correlation of the pixel motion field in the image sequence to figure out how the pixel is moving and thus the moving object [8]. Traditional methods for detecting moving vehicles in satellite video have inadequate precision and real-time performance; it is easily affected by satellites and

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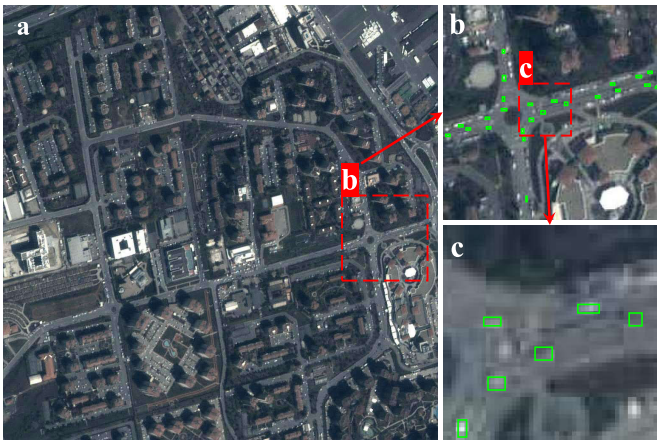


Fig. 1. Example of moving vehicle detection in VISO Jilin-1 satellite video dataset. (a) Area in a frame of the satellite video. (b) Partial enlarged image of (a). (c) Partial enlarged image of (b).

illumination; detection performance is dependent on manually designed features employing uniform parameters, and complex scenes make it difficult to achieve high detection accuracy.

In recent years, natural image object detection has made breakthroughs in multiple tasks [14], [19], [20], [21], [22], [23], [24], [25], providing new ideas and research methods for video object detection. Since the image object detection algorithm does not integrate the interframe information of video, the straightforward application of satellite video to the detection of moving vehicles is inefficient. Therefore, some researchers [11], [28] have used deep learning methods that incorporate spatio-temporal features. First, traditional approaches are utilized to extract difference information or construct background templates, and then deep learning methods are employed to further constrain the detection outputs in order to eliminate false-positives. This method relies heavily on the results of the background subtraction method and does not take advantage of the neural network's powerful feature extraction and expression capabilities. It is difficult to extract motion information effectively and has many false positives (FPs). Some researchers have designed a network model with continuous multiframe inputs based on object detection [12], [13] and tracking [27], [30] methods founded on multiframe information. However, the model has high computational complexity and low detection speed, making it challenging to meet real-time requirements.

Aiming at the above issues, performance involves two main issues: how to extract motion information effectively; and how to improve the detection speed of the model. Therefore, inspired by the multiframe differential detection method [12] and tracking method [27], [30], we propose the YOLO-HFDM satellite video motion vehicle detection network. In order to effectively extract information about moving vehicles, the historical frame difference module (HFDM) is designed, continuous frames are used as input to extract the spatio-temporal features of the image, and then the difference calculation is combined with the convolution operation so that the model can learn the different weights and obtain the motion fusion features of continuous frames. There is a lightweight feature extraction network using large-scale feature maps for model prediction, which aims to improve model accuracy and meet the detection speed. In order to improve the ability of the model to deal with complex scenarios, an improved multiscene mosaic (MSM) data augmentation strategy is used to increase the data complexity and make the network more robust. To prevent interframe information mixing, in the backbone and neck, group convolution [15] and group attention (GA) mechanism are used to improve the group feature expression ability of consecutive frames. The network structure in this article is shown in Fig. 2. The main contributions of this article are as follows.

1) In response to the problem of blurred moving vehicles in satellite videos and the difficulty distinguishing targets from similar interfering objects, we designed an HFDM. Using fixed interval multiframe images, motion information can be effectively extracted and fused with the current frame to enhance the current frame features, separating moving vehicles from the background using only historical information.

2) In order to improve model detection speed and ensure detection accuracy, we provide a useful bag of freebies for object detection tasks. Using improved data augmentation methods to enhance the robustness of the model; optimizing the backbone network to improve the feature extraction ability of the model; and using the GA mechanism to improve group feature expression.

3) In order to evaluate the proposed method, we conducted extensive experiments on a set of satellite videos and compared it with state-of-the-art algorithms. Our method can meet the real-time detection requirements while ensuring detection accuracy. The detection performance is superior to other classic vehicle detection methods.

The remainder of the article is structured as follows. Related works are presented in Section II. In Section III, we describe the satellite video moving vehicle detection model YOLO-HFDM in detail. Section IV describes the dataset, experimental configuration, and accuracy evaluation metrics, followed by extensive experiments and analysis. Section V concludes with conclusions and suggestions for future research.

II. Related Works

A. Traditional Object Detection Methods

The optical flow method is a prominent technique for object detection in conventional video. Li et al. [10] processed the image with the Canny edge operator, then calculated the displacement of each pair of edges employing the bidirectional optical flow method, constructed the displacement histogram for all matched edges, and finally matched the edge points of the moving object by using edge constraint to determine the moving object region. However, the optical flow method has a large amount of calculation and is sensitive to light, so it is unsuitable for scenes with significant light changes.

Some studies use background difference methods for moving vehicle detection. Shi et al. [7] proposed an improved Gaussian-based background subtractor (IPGBBS) algorithm that couples morphological closure operation and area filtering to improve detection accuracy. Ahmadi et al. [18] detected moving objects utilizing a background subtraction method and per-frame Otsu thresholding. However, the background

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difference method is more sensitive to occlusion, illumination changes, and background jitter in the scene, especially in the case of background jitter, which can generate many pseudo-moving targets.

As to the frame difference method, Chen et al. [6] proposed a novel moving vehicle detection approach using adaptive motion separation and difference accumulated trajectory. Using the frame difference method to perform adaptive motion separation and obtain a differential binary image based on the scale invariance of moving vehicles in satellite video, they suppressed pseudo-moving targets based on differential trajectory accumulation. This method suppresses false alarms and allows detection in real-time. Shu et al. [5] proposed a local enhancement fusion Gaussian mixture model and improved a three-frame difference detection framework, namely, E-GMMTFD. In the E-GMMTFD framework, the local contrast enhancement method improves the contrast between the target and the background, fusing Gaussian background models with three-frame difference methods to reduce false detections caused by illumination and background motion. The frame difference method is simple and fast. However, it has an inadequate detection effect for slow-moving targets, is vulnerable to environmental noise, and is easy to produce "holes" in moving targets.

B. Object Detection Methods Based on Deep Learning

Object detection methods in the field of computer vision have achieved state-of-the-art detection performance in the field of natural images. The classical object detection networks can be divided into the following two categories. 1) Two-stage detection network. For example, R-CNN [19], Fast R-CNN [20], and Faster R-CNN [21]. 2) One-stage detection network. For example, SSD [22], YOLOv5 [14], YOLOX [23], YOLOv8 [24], and CenterNet [25]. Among them, the detection speed of the one-stage model is faster than that of the two-stage model, but the detection accuracy is slightly lower than that of the two-stage model.

For moving object detection in satellite video, some studies have improved the relevant algorithms [47], [48]. LaLonde et al. [1] proposed a two-stage spatio-temporal CNN that first employs a set of extremely large video frames, combines motion and appearance information in a convolutional architecture to extract candidate motion regions, and then determines motion targets by heat map estimation. Chen et al. [11] proposed a lightweight KNN+LeNet5 framework that employs a K-nearest neighbor (KNN) background subtraction model to generate preliminary detection results followed by a shallow convolutional neural network to suppress false alarm objects. The strategy is simple, but it heavily relies on the results of background subtraction, it cannot utilize the powerful feature extract ability of natural networks and has poor real-time transmission performance with only 2.9 FPS.

For the processing of multiframe information, some scholars have also adopted tracking-based approaches. Du et al. [27] proposed a method that combines kernel correlation filter (KCF) and three-frame difference to improve the tracking performance of small targets. However, this method is sensitive to light and noise, and its time performance needs to be improved. Wu et al. [30] proposed a multitarget tracking method that considers both slow and motion features and designed a non-maximum suppression (NMS) module to assist in the target detection section, but there are still static false detections. Ao et al. [59] used three frame differences to obtain binary difference images and then utilized a discrimination algorithm based on multiple morphological cues and track association to detect tiny moving vehicles. However, it still performs inaccurately in weak feature texture images.

Some scholars have introduced tracking ideas into detection networks, Feng et al. [12] proposed a cross-frame keypoint-based detection network (CKDNet) based on CornerNet, which extracts interframe information via the designed cross-frame module to enhance the detection capability of the model using interframe information. The strategy combines the idea of frame differential with deep learning, which has a good performance, but at the cost of lower precision and significantly higher recall, and it does not work well in real-time. Xiao et al. [13] designed a combined dynamic and static network for satellite video moving vehicle detection by extracting contextual information from static single frames and dynamic motion cues from continuous frames. With direct input of multiple frames, however, information from different frames can get mixed up, forcing the use of 3-D convolution, which has high computation consumption and does not meet real-time requirements. Pi et al. [33] facilitated the extraction of semantic features by integrating motion information from adjacent frames and combining Transformer to refine for key point estimation and scale prediction, the model can identify actual moving objects well and suppress interference from stationary objects or backgrounds. However, the use of 2K frames to generate temporal context information and the highly computationally consuming nature of the method limits its applicability to real-world applications.

The mentioned studies utilize motion information from multiple frames and combine deep learning techniques with satellite video characteristics, which makes it easier to find moving objects in satellite video. However, there are still some problems: The majority of studies use image information after the current frame, which is unsatisfactory for real-time detection; the direct fusion of interframe differential features may have the problem of interframe information intermixing, which affects detection accuracy; only use simple data augmentation methods, which does not fully use the multiscene information in the dataset and face challenges in detecting moving vehicles in complex scenes; the model requires much computation, and the detection speed is brutal to meet the real-time requirements. Therefore, we improve a lightweight YOLO-HFDM network model based on the YOLOv5 model, and more details will be introduced in Section III.

III. Proposed Approach

The network model structure of this article is shown in Fig. 2. First, the data complexity of continuous multiframe is improved by MSM data augmentation, the obtained continuous

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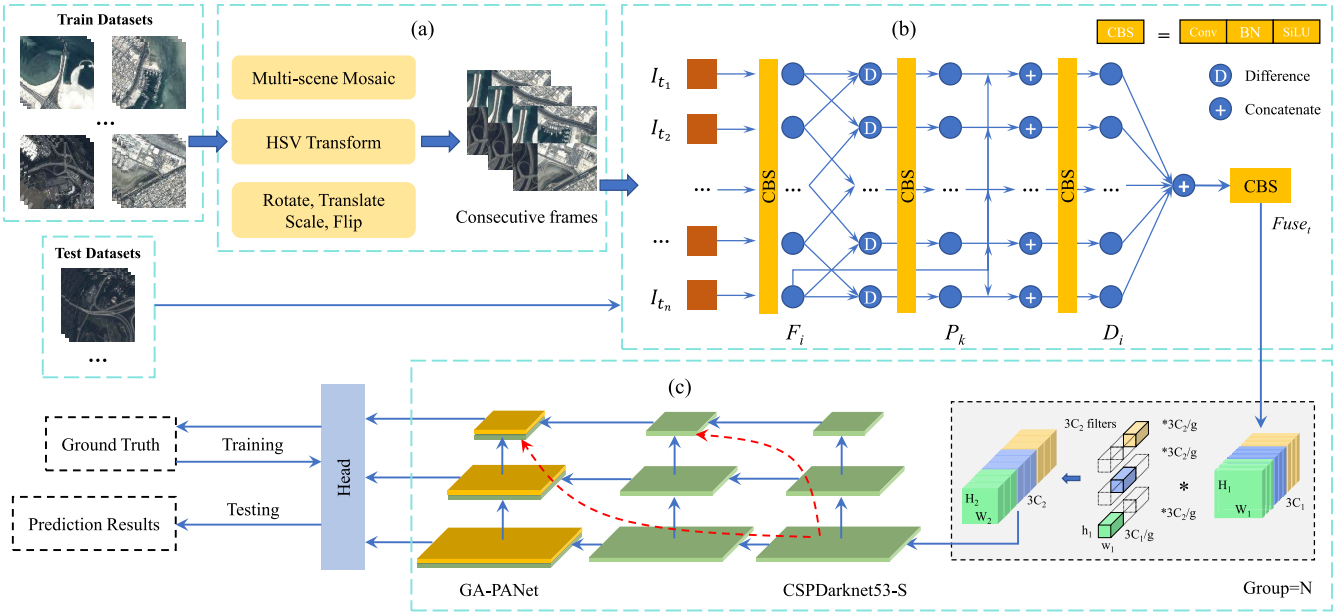


Fig. 2. Satellite video moving vehicle detection network structure. It contains four main components as follows. 1) Use improved data augmentation strategies to increase data complexity. 2) The HFDM is used to extract the motion information of consecutive frames. 3) Use improved backbone and neck for feature extraction and feature fusion. 4) Use the detection head to output the final detection results. (a) Data augmentation. (b) HFDM. (c) Backbone+neck.

frame images are input into HFDM, and the differential features of the current frame and the historical frames are fused to obtain the motion fusion features of the continuous frames. The obtained motion fusion features are input into the feature extraction network with group convolution, CSPDarknet53-S is used as a backbone, and group $= N$ is set for group convolution [15]. In the neck, the GA-path aggregation network (PANet) structure is used to obtain the multiscale information of motion fusion features and improve the grouped feature representation, at the same time, reducing the down-sampling rate of feature extraction to improve the detection performance of small targets. Finally, the deep multiscale fusion features are transferred to the detection head to generate the final detection results.

A. MSM Data Augmentation

The data augmentation strategy can effectively increase the diversity of the dataset and make the model more robust in the face of complex scenes [29], [32], [50]. Common data augmentation strategies for object detection tasks include photometric and geometric distortions, which can effectively expand the diversity of the dataset. However, they are pixel-wise adjustments that retain all the information in the image and have limited data expansion capability, difficulty combining the characteristics of different satellite video data, and the scene information is relatively singular.

As shown in Fig. 3, some researchers have improved data augmentation methods based on simulated image occlusion and image stitching. Such as mixup [31], random erasing [51], cutout [52], and grid mask [53], which may be hard to apply to blurrier satellite videos effectively. Smaller moving vehicles have almost no structural texture features, and mask-based data augmentation methods will only generate noninformative pixels, which is not conducive to model training. Cutmix [54] and mosaic [32] use multiple images for stitching, which can enrich the data diversity and reduce the demand for batch size. However, cutmix combines only two images with limited data complexity. As for mosaic, the stitched four images have the same pixel-level data augmentation approach and single-scale information. Consequently, based on mosaic, we adopt an improved MSM data augmentation approach to improve the spatial scale information in a single image and enrich the data diversity.

In order to increase the complexity of the scenario, we use image stitching, while needing to ensure the uniformity of the enhancement over multiple consecutive frames. As for a single-frame image, first, create a new blank template with size S, where S is equal to the original image size, and then set the random center (x, y) to get the size of the four regions as $\text{Size}_{\text{bbox}} = \{(x, y), (s-x, y), (x, s-y), (s-x, s-y)\}$. Then, four frames of four videos are randomly selected from the dataset, positioned, cropped, and scaled randomly, and finally stitched into a complete multiscene continuous multiframe image, where scale $> = 1$ is to prevent the targets from becoming smaller and causing detection difficulties. Finally, although MSM data augmentation can improve the richness of the data, it changes the label distribution of the training samples. For the model to complete convergence in natural scenes, the data augmentation early stops strategy is set, i.e., data augmentation is turned off in the last N epochs. Multiscene stitching of consecutive frames is achieved utilizing improved MSM data augmentation, which enriches the scene scale information of single-frame images without generating noninformative pixels and scene distortion affecting model training.

B. Historical Frame Differential Module

In order to introduce the frame difference method into the convolutional neural network, Feng et al. [12] and Xiao et al. [13] proposed a cross-frame module. However,

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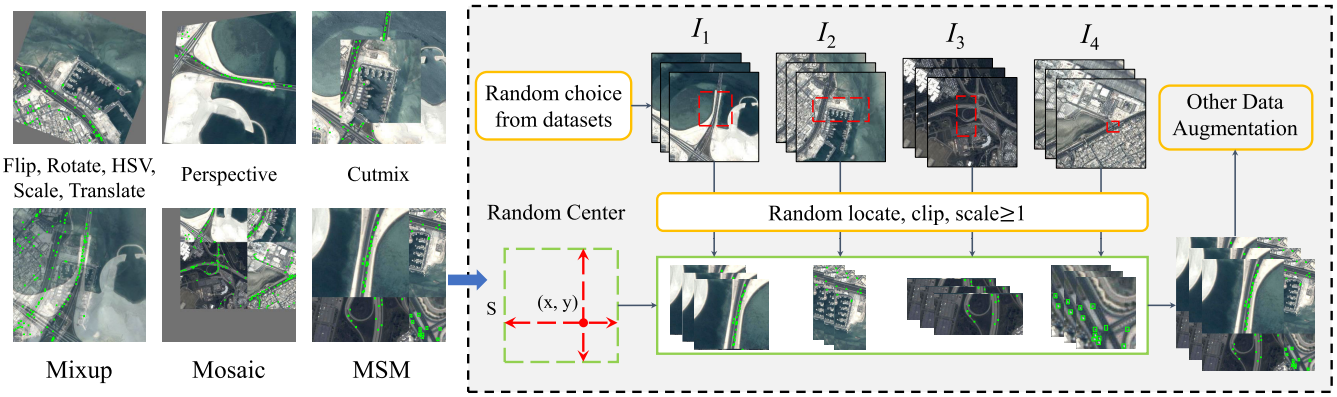


Fig. 3. Various methods of data augmentation and MSM data augmentation flowchart. I_1–I_4 are three consecutive frames of any four videos in the training dataset.

image data after the current frame is used, which cannot achieve the real sense of real-time, and the direct fusion of the three frames' differential features suffers from the problem of interframe information intermixing. In this article, we design the HFDM and the structure is shown in Fig. 2. For the continuous input frames $V = \{I_{t_1}, I_{t_2}, \ldots, I_{t_n}\}$, in which I_{t_n} indicates the current frame and the others indicate historical frames and Δt represents the interval between adjacent frames. The conv, batch normalization, and sigmoid-weighted linear unit (SiLU) (CBS) are performed on each frame individually to obtain the shallow spatial features, i.e., $F = \{F_{t_1}, F_{t_2}, \ldots, F_{t_n}\}$. By differential and convolution, the differential temporal features $P = \{P_{t_1}, P_{t_2}, \ldots, P_{t_n}\}$ are derived from the shallow spatial features of consecutive frames. The detailed equation is illustrated below

$$\mathrm{SiLU}(x) = \frac{x}{(1+e^{-x})} \tag{1}$$

$$F_i = \mathrm{CBS}_1(I_i), \quad (i \in (t_1, t_2, \ldots, t_n)) \tag{2}$$

$$P_k = \mathrm{CBS}_2\big(\mathrm{AbsDiff}\big(F_i, F_j\big)\big) \quad (i, j, i \neq j, k \in (t_1, t_2, \ldots, t_n)) \tag{3}$$

where x is the input to the hidden unit; conv in CBS_1 is the 2-D convolution with $K = 6$, $S = 2$, and $P = 2$; conv in CBS_2 is the 2-D convolution with $K = 3$, $S = 1$, and $P = 1$; AbsDiff$(\cdot)$ represents absolute difference; and i and k represent any one of the input multiple frames.

The differential temporal feature information $P = \{P_{t_1}, P_{t_2}, \ldots, P_{t_n}\}$ and the shallow spatial features F_{t_n} are fused to obtain fused spatio-temporal information features, i.e., $D = \{D_{t_1}, D_{t_2}, \ldots, D_{t_n}\}$, highlighting the feature information of the moving object in consecutive frames and reinforcing the current frame features, finally concatenate in the channel dimension to obtain the final fused features Fuse_{t_n}, which is defined as

$$D_i = \mathrm{CBS}_3\big(\mathrm{Concat}\big(F_{t_n}, P_i\big)\big), \quad (i \in (t_1, t_2, \ldots, t_n)) \tag{4}$$

$$\mathrm{Fuse}_{t_n} = \mathrm{CBS}_4\big(\mathrm{Concat}\big(\big(D_{t_1}, D_{t_2}, \ldots, D_{t_n}\big)\big)\big) \tag{5}$$

where conv in CBS_3 is the 2-D convolution with $K = 3$, $S = 1$, and $P = 1$; conv in CBS_4 is the 2-D convolution with $K = 3$, $S = 1$, $P = 1$, and $G = n$; and Concat represents the connection of feature maps in the channel dimension.

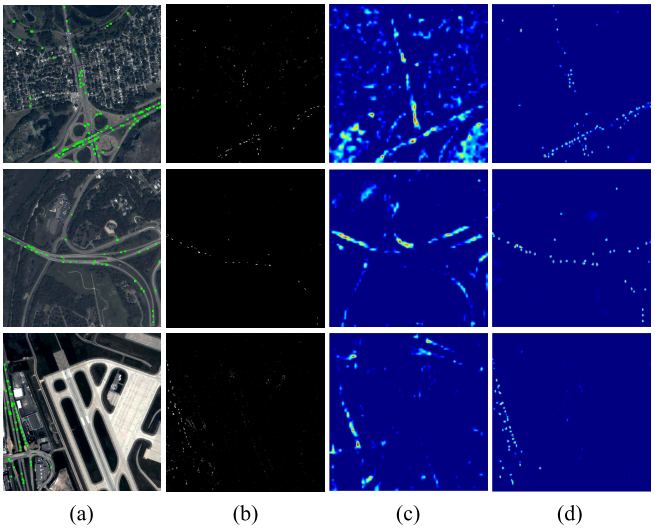


Fig. 4. Visual comparison of different methods. (a) Examples of the JiLin-1 satellite images. (b) Background template of the traditional method. (c) w/o HFDM. (d) w/ HFDM.

In a continuous video sequence, if there is a significant interval between adjacent frames, there will be a particular background offset, making it challenging to extract motion features effectively. On the contrary, it is difficult to capture motion information. Therefore, we designed differential modules with fixed intervals to ensure the differential features have similar amounts of motion and better reflect the moving target. By default, the parameters $n = 3$ and $\Delta t = 2$ are set.

We use current and historical frames as inputs to the network, where historical information is only used to extract motion features, separating moving vehicles from the background. Therefore, we only use the moving vehicle in the current frame as the truth label.

Since the background of adjacent frames in satellite video hardly moves, the change information between two frames can be obtained by differencing, which contains moving vehicles, building edges, and noise, as shown in Fig. 4(b), and it is still difficult to distinguish moving targets effectively. CNN-based neural networks can obtain semantic information from images and indicate potential vehicle targets from the background. Due to the lack of temporal information, it is difficult to

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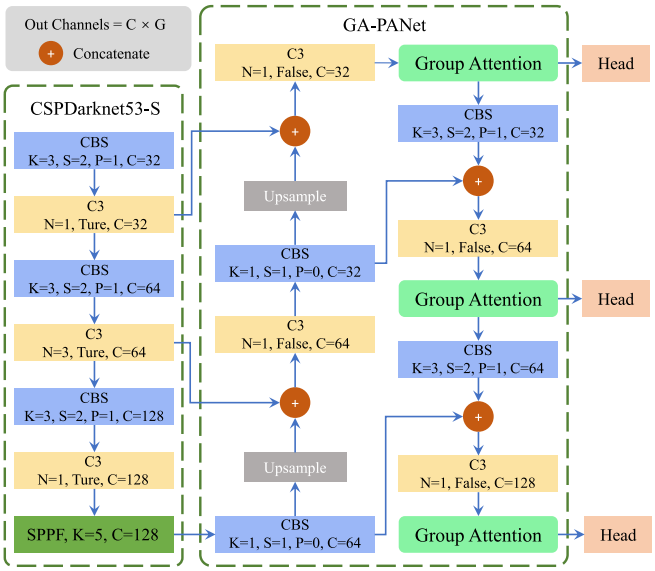


Fig. 5. CSPDarknet53-S and GA-PANet network architecture.

effectively distinguish between motion and stillness, as shown in Fig. 4(c). As for HFDM, it combines differential temporal information and spatial semantic information, extracts change features using differential temporal information, and suppresses building edges and noise using spatial semantic information. Therefore, it can effectively separate moving vehicles from the background, as shown in Fig. 4(d).

C. Object Detection Network

CSPDarknet53 [34], as a classical feature extraction network, introduces the cross-stage partial (CSP) structure based on DarkNet53, which can enhance the learning ability of convolutional neural networks and reduce the required memory bandwidth, so CSPDarknet53 is used as a backbone in this article. Also, since the moving vehicle in satellite video occupies only a dozen pixels, we reduced the downsampling rate of CSPDarknet53 and used large-scale feature maps for prediction. The optimized network structure is named CSPDarknet53-S. The PANet [55] structure is used in the neck, bottom-up feature transfer to complement the FPN [35] structure, which is beneficial to enhance the extraction ability of small targets.

In the motion vehicle detection of satellite video, we use HFDM to obtain the spatio-temporal feature information of multiframes, so in the network, to ensure the integrity and independence of the information of consecutive frames, the standard convolution in CBS and C3 is changed to group convolution with group $= G$. Among them, G is equal to different input frames, so the number of output channels varies for different inputs. The number of output channels for each layer in the network can be represented as $C \times G$, where C for each layer is taken as shown in Fig. 5.

Attention mechanism has become a vital module to improve CNN feature expression, and in general, the commonly used attention mechanisms have spatial attention mechanisms and channel attention mechanisms. In the neck, this article has

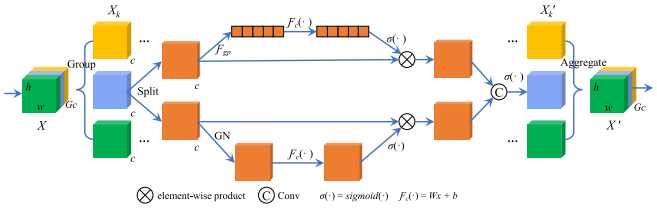


Fig. 6. Overview of the proposed GA module.

been improved based on shuffle attention (SA) [56] and named as GA mechanism to improve the expression of grouped features with multiple consecutive frames. Suppose the input features, i.e., $X \in R^{G\times C\times W\times H}$, where C, W, and H are the number of channels, width, and height, respectively, and divide X into G groups according to channel dimension, i.e., $X = [X_1, \ldots, X_G]$, $X_k = R^{C\times W\times H}$, and then divide X_k into two branches, i.e., X_{k_1}, $X_{k_2} \in R^{C\times W\times H}$, the spatial and channel attention mechanisms are used to improve the feature representation of spatial and channel, respectively, and two enhanced feature maps are obtained, i.e., X'_{k_1}, $X'_{k_2} \in R^{C\times W\times H}$. Finally, the feature maps of the two branches are fused using 2-D convolutional computation and aggregated between groups to obtain the grouped attention feature maps, i.e., $X' \in R^{G\times C\times W\times H}$. The GA module is shown in Fig. 6, and the network framework is shown in Fig. 5.

D. Loss Function

After YOLO-HFDM, the preliminary detection results are obtained, and the predictions are made for the bounding boxes, confidence, and category, respectively. To improve the detection results of the model, the overall loss function of the model is as follows:

$$L = \lambda_1 L_{\text{cls}} + \lambda_2 L_{\text{obj}} + \lambda_3 L_{\text{bbox}} \tag{6}$$

where L_{cls}, L_{obj}, and L_{bbox} are classification loss, confidence loss, and bounding boxes loss, respectively, and λ_1, λ_2, and λ_3 are different weights factors. To better balance the losses and make the bounding boxes fit better, we set λ_1, λ_2, and λ_3 to 1, 1, and 0.05, respectively.

Due to the extremely uneven distribution of positive and negative samples in the prediction bounding boxes, the category and confidence loss for detecting moving vehicles from satellite video is changed to focal loss [17], which is defined as

$$\text{FL}(p_t) = -(1-p_t)^{\gamma}\log(p_t) \tag{7}$$

$$p_t = \begin{cases} p, & \text{if } y = 1 \\ 1-p, & \text{otherwise} \end{cases} \tag{8}$$

where y is the prediction category; p is the prediction probability of the model category when $y = 1$; and γ is a constant.

In the bounding box loss, considering that the moving vehicle is small, basically occupying only a few or a dozen pixels, it is difficult to effectively fit the truth boundary using only the intersection of union (IoU) indicator. In this article, we use efficient intersection over union (EIoU) [16] as the

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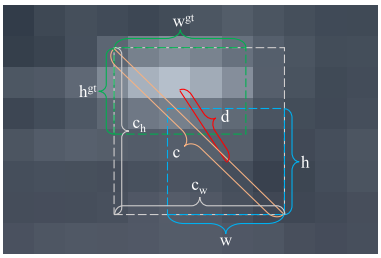


Fig. 7. Example diagram of bounding boxes.

bounding boxes loss, which integrates the IoU, distance, and aspect ratio and can effectively improve the convergence speed and regression accuracy of the bounding boxes, which is defined as

$$L_{\text{EIoU}} = L_{\text{IoU}} + L_{\text{dis}} + L_{\text{asp}} = 1 - \text{IoU} + \frac{\rho^2(b, b^{gt})}{c^2} + \frac{\rho^2(w, w^{gt})}{C_w^2} + \frac{\rho^2(h, h^{gt})}{C_h^2} \tag{9}$$

where C_w and C_h are the width and height of the smallest enclosing box covering the two boxes, other symbols are shown in Fig. 7; $\rho^2(b, b^{gt})$ is the square of the distance between the center point of the prediction bounding boxes and the truth bounding boxes.

IV. Experimental Results and Analysis

A. Experimental Dataset

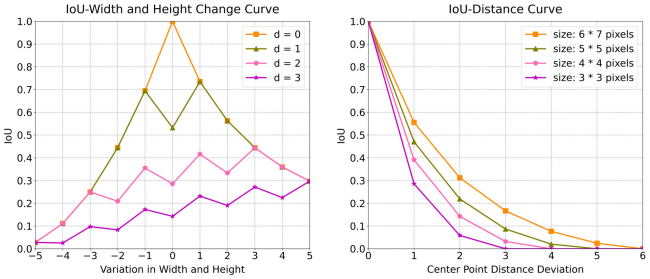
The training dataset utilized in this article is the public dataset VISO [36], which consists of 21 videos with 6716 frames and a 1024 × 1024 image resolution. Two datasets are selected as test datasets to verify the performance of the model. The test dataset I is the VISO Jilin-1 satellite video dataset, which contains six videos with 600 frames and an image size of 1024 × 1024 pixels. The test dataset II is captured by Jilin1-3 over Dubai, United Arab Emirates, with three videos of 450 frames each and a 640 × 640 image resolution.

B. Experimental Setups

The experimental environment in this article: the operating system is CentOS 7; the CPU is Inter[1] Xeon[1] E5-2640 v4 2.40 GHz; the GPU is NVIDIA Tesla P40, and the deep learning framework is the PyTorch environment.

The following parameters have been set up for this article: the gradient descent optimization algorithm is the Adam [37] algorithm with the momentum of 0.937, weight decay of 0.0005; the initial learning rate of 0.001, learning rate decay with CosineAnnealing, minimum learning rate ratio of 0.2; batch size of 8, number of training epochs of 50; learning rate of 0.1 and momentum of 0.8 in the first three epochs of the warm-up [38] training process, data augmentation early stop parameter $N = 5$; use focal loss to balance the category and confidence loss and set the hyperparameter γ to 1.5; set the parameters of HFDM $n = 3$, $\Delta t = 2$.

[1]Registered trademark.



(a) (b)

Fig. 8. (a) IoU and center point distance change curves. (b) IoU and bounding boxes size change curves.

C. Evaluation Metric

The following equations are used to quantitatively analyze the method provided in this article, considering precision, recall, and $F1$ scores as quantitative evaluation measures

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{10}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{11}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{12}$$

where true positive (TP) is the correct detection; FP indicates the false alarms; and false negative (FN) indicates the missed detection. The prediction objects are judged to belong to TP, FP, and FN based on the IoU of the prediction bounding boxes and the truth bounding boxes. As can be seen from the IoU-deviation graph in Fig. 8, due to the tiny target of the satellite video motion vehicle, slight bounding box differences and center point offsets can seriously affect the IoU [49], the conditions for determining it as TP are too strict. So the IoU threshold is set to 0.3 in this article, which can satisfy moving vehicle localization and bounding box fitting.

Finally, FPS is utilized as the speed assessment index with the following equation to test the model detection speed

$$\text{FPS} = \frac{1}{T} \tag{13}$$

where T is the inference time of a single image.

D. Performance Evaluations

In this section, to verify the superiority of the method proposed in this article, existing state-of-the-art methods in object detection are compared, including traditional methods (Vibe [39], T2FGMM_UM [40], DPAdaptiveMedian [41], MOGv2 [42], and D&T [59]) and deep learning methods (CenterNet-ResNet50 [25], YOLOv5 [14], YOLOX [23], and DSFNet [13]). All methods were retrained on the same dataset used in this article. Fig. 9 and Table I show the qualitative and quantitative evaluations of different methods on the VISO dataset.

From the qualitative results in Fig. 9, it can be seen that Vibe [39], T2FGMM_UM [40], MOGv2 [42], and DPAdaptiveMedian [41] have inferior detection results in dense vehicles areas. It is difficult to accurately locate

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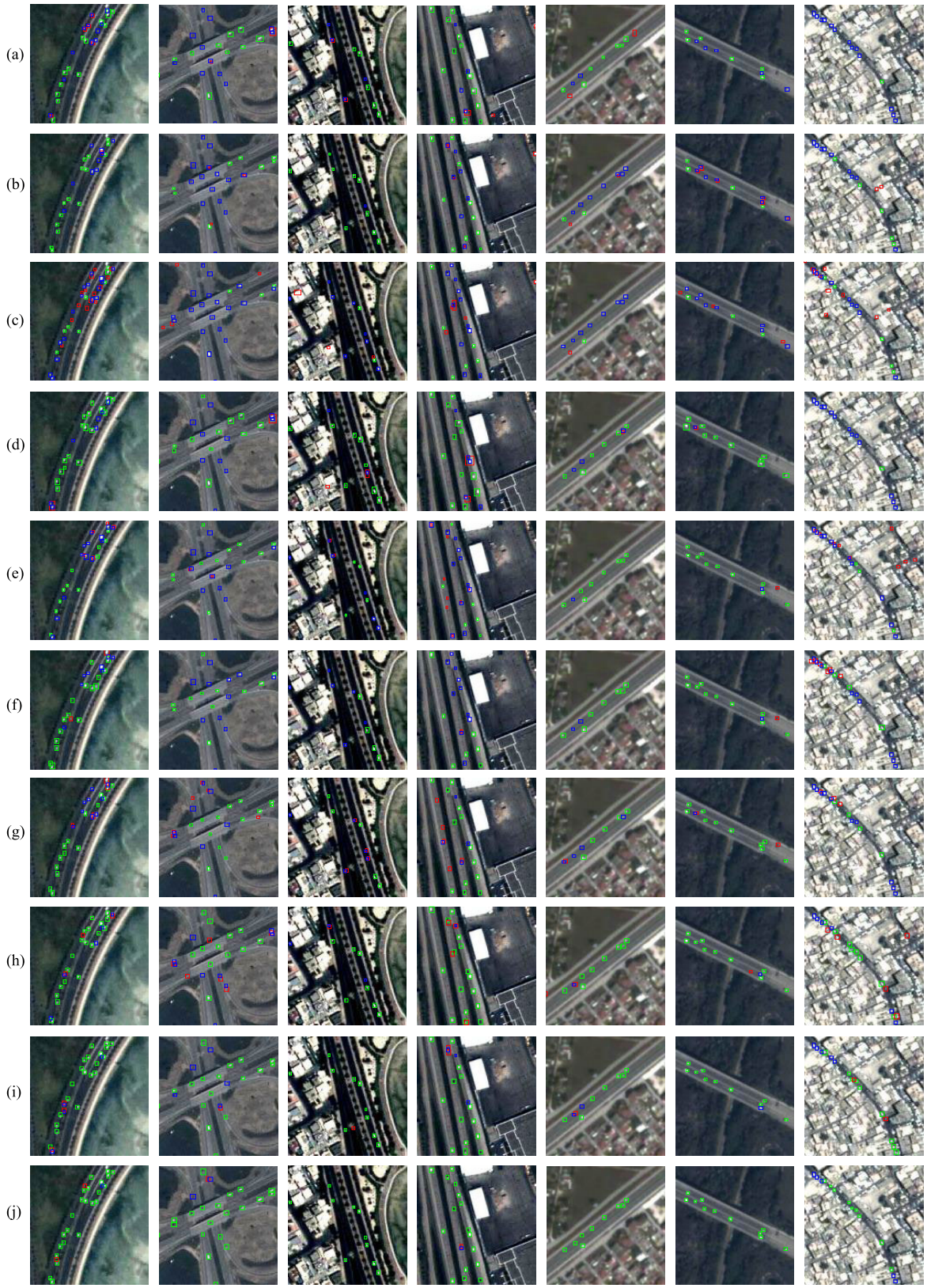


Fig. 9. Qualitative results of different comparison methods in test dataset I. The first six columns of scenes are from six different test datasets I; the seventh column is from video_1. (a) Vibe [39]. (b) T2FGMM_UM [40]. (c) DPAdaptiveMedian [41]. (d) MOGv2 [42]. (e) CenterNet-ResNet50 [25]. (f) YOLOv5 [14]. (g) YOLOX [23]. (h) D&T [59]. (i) DSFNet [13]. (j) YOLO-HFDM (Ours). Green boxes: TP. Red boxes: FN. Blue boxes: FP.

vehicles because the traditional methods rely on background modeling, which also suppresses vehicles with low contrast in order to suppress noise points. The image-level object detection algorithms, such as YOLOv5 [14], CenterNet [25],

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TABLE I

QUANTITATIVE EVALUATION RESULTS OF DIFFERENT COMPARISON METHODS IN TEST DATASET I, THE BEST RESULTS ARE SHOWN IN **RED**, AND THE SECOND BEST RESULTS ARE SHOWN IN **BLUE**

Methods	Video_1			Video_2			Video_3		
	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)
Vibe [39]	53.82	46.40	49.84	70.31	42.90	53.29	49.37	51.94	50.60
T2FGMM_UM [40]	73.67	41.08	52.75	57.14	42.00	48.41	63.28	62.53	62.90
DPAdaptiveMedian [41]	20.92	28.94	24.28	42.82	15.92	23.22	35.74	15.83	21.94
MOGv2 [42]	41.16	40.60	40.88	42.06	48.83	45.19	15.14	48.00	23.02
CenterNet-ResNet50 [25]	51.27	30.23	38.03	73.51	47.78	57.92	41.43	17.43	24.54
YOLOv5 [14]	78.62	50.22	61.29	81.27	52.04	63.45	57.07	29.01	38.47
YOLOX [23]	76.49	42.82	54.91	68.45	57.83	62.69	79.97	37.45	51.01
D&T [59]	76.59	46.58	57.93	73.78	58.94	65.53	70.10	46.12	55.64
DSFNet [13]	**87.74**	**84.12**	**85.89**	**92.37**	**82.30**	**87.04**	**87.34**	**83.18**	**85.21**
YOLO-HFDM (Ours)	**91.29**	**83.78**	**87.38**	**93.90**	**83.04**	**88.14**	**92.70**	**82.89**	**87.52**

Methods	Video_4			Video_5			Video_6			Average		
	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)
Vibe [39]	53.55	50.96	52.22	34.74	55.46	42.72	53.05	43.60	47.86	52.26	48.21	50.15
T2FGMM_UM [40]	40.60	55.60	46.93	35.74	30.81	33.09	48.29	19.52	27.80	58.75	43.46	49.96
DPAdaptiveMedian [41]	35.47	27.25	30.82	37.47	11.71	17.84	26.88	15.87	19.95	25.82	22.09	23.81
MOGv2 [42]	25.64	57.45	35.46	33.42	43.95	37.97	69.74	54.30	61.06	31.06	46.31	37.18
CenterNet-ResNet50 [25]	41.70	37.01	39.22	86.18	51.84	64.74	78.10	83.10	80.52	59.03	38.11	46.32
YOLOv5 [14]	76.87	44.23	56.16	66.04	58.00	61.67	79.59	85.41	82.40	75.17	47.43	58.16
YOLOX [23]	68.46	68.11	68.29	76.21	46.17	57.50	82.95	73.74	78.07	72.02	45.61	55.80
D&T [59]	64.32	57.80	60.88	73.72	49.58	59.48	55.75	51.60	53.59	70.91	50.40	58.92
DSFNet [13]	**83.78**	**84.26**	**84.02**	**83.45**	**73.49**	**78.16**	**85.87**	**92.22**	**88.93**	**86.62**	**82.73**	**84.63**
YOLO-HFDM (Ours)	**87.30**	**86.76**	**87.03**	**92.41**	**80.14**	**85.84**	**91.97**	**93.54**	**92.74**	**93.35**	**82.83**	**87.78**

and YOLOX [23], do not combine spatio-temporal features between frames, are ineffective in detecting ambiguous vehicles, and cannot differentiate between stationary and moving vehicles as a result. D&T [59] uses trajectory information to assist detection but performs inadequately on vehicles with weaker texture features. Although the DSFNet [13] method combines interframe information and achieves high detection performance, some low-contrast vehicles still have missed detections and insufficient bounding box fitting. The YOLO-HFDM proposed in this article has the best overall performance on each video test dataset I, and the robustness of the model is high enough to overcome complicated scenarios. The full combination of interframe time information helps to improve the performance of moving vehicle detection in satellite video.

From the quantitative results in Table I, we can see that YOLO-HFDM has the highest $F1$ scores on each video dataset, and the average $F1$ score is 87.78%, which is due to the fact that YOLO-HFDM fully combines spatio-temporal information to identify motion target information from multiframe images and improves the model performance to cope with complex scenes utilizing MSM data augmentation. Compared with YOLOv5 [14], CenterNet [25], and YOLOX [23], the accuracy of the proposed YOLO-HFDM improves significantly in more complex videos. The average detection accuracy of DSFNet [13] ranks second, indicating that combining static single-frame and dynamic multiframe methods can also effectively improve the detection accuracy of moving vehicles in satellite videos. For traditional methods, setting fixed parameters makes it difficult to cope with different scenarios effectively. D&T [59] integrates trajectory information to achieve relatively high detection accuracy, indicating that tracking-based cross-frame features can improve detection performance. The method proposed in this article can overcome the problem of detection in complex scenes and achieve high detection accuracy without setting artificial parameters.

In order to compare the individual video detection performance in-depth, we designed to compare the performance of different methods on a single video with different frames. The comparison results are shown in Fig. 10. It can be seen that since the traditional methods are based on background modeling, the background information is not yet established at the first frame, so the $F1$ scores of the detection results at the first frame are all 0, and it takes a longer time to reach the best accuracy of the model, but there are still a lot of false alarms and miss detection. The performance of DSFNet [13] is more stable on all frames, which is due to the fact that DSFNet [13] combines the features of static and dynamic streams and achieves a higher accuracy throughout the video frames. As for CenterNet [25], YOLOv5 [14], and YOLOX [23], since such methods are single-frame, image-level object detection methods that do not fully use interframe information have lower precision and fluctuate greatly in the detection process. The detection of the method YOLO-HFDM in this article also relies on information from historical frames. The first frame detection cannot utilize historical information, so the accuracy of the first frame detection is also relatively low, but after the third frame, YOLO-HFDM can stably achieve the best detection performance.

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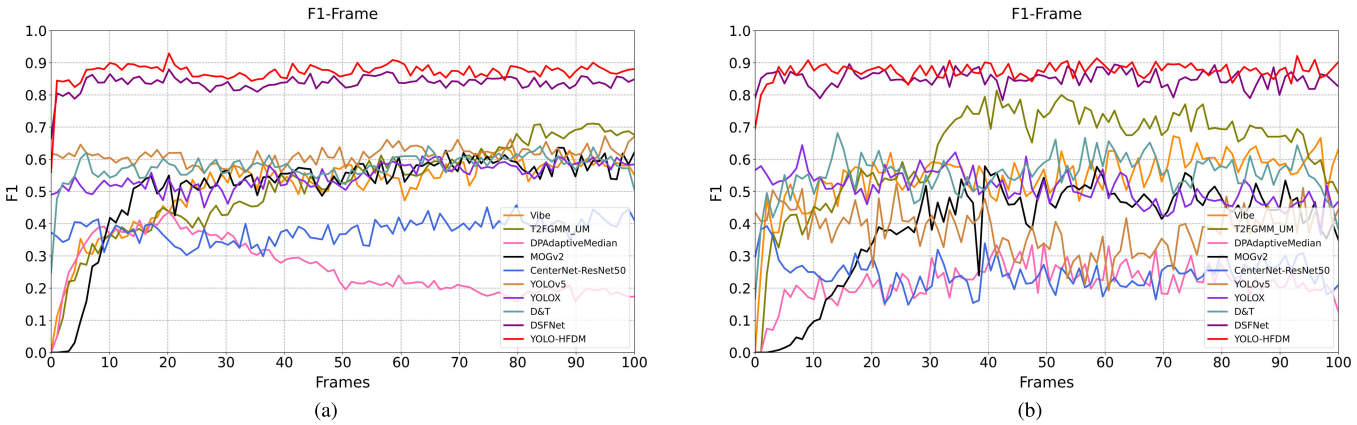


Fig. 10. Detection results of single frames on (a) video_1 and (b) video_3 for different comparison methods.

TABLE II
PARAMETERS AND SPEED OF DIFFERENT COMPARISON METHODS

Methods	F1(%)	Time Cost(ms)	FPS	Params(M)
Vibe [39]	50.15	14.86	67	-
T2FGMM_UM [40]	49.96	48.58	21	-
DPAdaptiveMedian [41]	23.81	19.51	51	-
MOGv2 [42]	37.18	25.64	39	-
CenterNet-ResNet50 [25]	46.32	27.37	37	32.67
YOLOv5 [14]	58.16	22.49	44	7.01
YOLOX [23]	55.80	24.80	40	8.94
D&T [59]	58.92	113.57	9	-
DSFNet [13]	84.63	336.65	3	6.71
YOLO-HFDM (Ours)	87.78	47.62	21	3.18

The real-time of the detection algorithm is also crucial in practical applications, therefore, we evaluate the detection speed, inference time, and model parameters, as shown in Table II. As the traditional method does not require NMS operation and generates detection results directly by background subtraction or background difference, the short inference time gives the model a higher detection speed. However, due to the unified superparameters, it is difficult to adapt to diverse scenes, so the detection accuracy is inferior. As for T2FGMM_UM [40], due to the computationally complex GMM, it takes up more computing time, so the detection speed is lower than other traditional methods. D&T [59] uses three frame differences and track associations, and the FPS is 9, with a slower detection speed. For CenterNet [25], YOLOv5 [14], and YOLOX [23] have general performance in speed and accuracy, allowing them to meet real-time demand, but they cannot achieve high detection accuracy. DSFNet [13] uses five consecutive frames as input and 3D convolution, with higher computational complexity and slower inference time. The $F1$ score of the method proposed in this article is 87.78%, which achieves the highest detection accuracy among the compared methods, and the FPS is 21, with a slower detection speed than other methods, but it can meet the demand for real-time detection. In addition, the backbone and neck of this article are lighter, the model parameters are only 3.18M, and the inference time is lengthened because the model input is transformed from a single frame to a continuous multiframe. YOLO-HFDM balances the need for speed and accuracy to meet the real-time performance needs for satellite video moving vehicle detection.

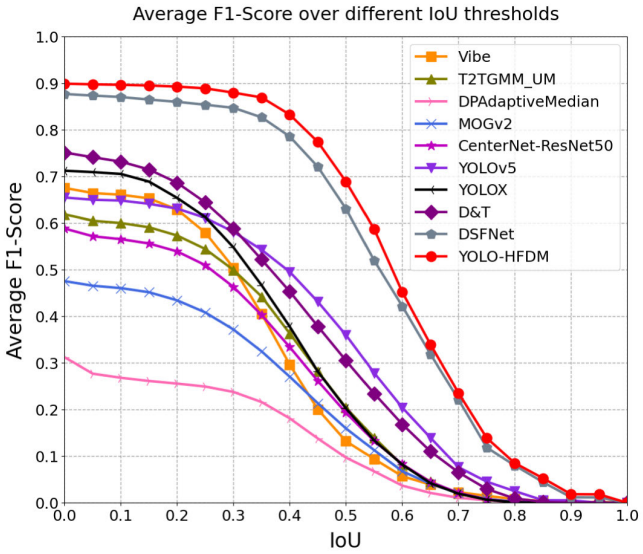


Fig. 11. $F1$-scores curves of the different comparison methods in different IoU thresholds.

To further test the model performance, the $F1$ scores of the model were evaluated at different IoU thresholds, and the results are shown in Fig. 11. Compared with other methods, YOLO-HFDM can maintain the best performance as the IoU threshold increases. When the IoU threshold increases to 0.5, YOLO-HFDM can still achieve $F1$ scores of 0.69, while the average $F1$ scores of some of the compared methods decrease more rapidly. This is because traditional methods lack robustness, have many noise points, and have insufficient boundary fitting. Deep learning methods such as YOLOv5 [14], YOLOX [23], and CenterNet [25] are image-level object detection networks that do not combine temporal information from the video, making it difficult to distinguish between dynamic and static information, and are incapable of accurately predicting the location of the objects. D&T [59] combines tracking methods to achieve high $F1$ at low IoU, but $F1$ decreases rapidly with increasing IoU. In this article, the

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TABLE III

ACCURACY OF DIFFERENT COMPARISON METHODS ON THE TEST DATASET II, THE BEST RESULTS ARE SHOWN IN **RED**, AND THE SECOND BEST RESULTS ARE SHOWN IN **BLUE**

Methods	Video_1			Video_2			Video_3			Average		
	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)
Vibe [39]	53.63	69.18	60.42	35.78	46.13	40.30	49.28	55.08	52.02	46.23	56.80	50.91
T2FGMM_UM [40]	52.75	42.51	47.08	47.53	22.91	30.92	47.12	29.49	36.28	49.13	31.64	38.09
DPAdaptiveMedian [41]	15.89	30.71	20.94	7.84	22.01	11.56	16.63	23.16	19.36	13.45	25.29	17.29
MOGv2 [42]	51.52	70.05	59.37	45.68	50.17	47.82	42.98	56.32	48.75	46.90	58.90	52.22
CenterNet-ResNet50 [25]	73.26	75.28	74.26	80.34	49.31	61.11	64.32	63.74	64.03	72.64	62.78	66.46
YOLOv5 [14]	72.90	78.66	75.67	73.11	64.01	68.26	71.43	66.00	68.61	72.50	69.61	71.03
YOLOX [23]	80.18	66.42	72.65	**86.59**	46.26	60.30	**77.72**	57.63	66.19	80.84	56.78	66.71
D&T [59]	75.28	79.72	77.43	83.74	62.14	71.34	71.19	63.95	67.38	76.29	68.67	72.28
DSFNet [13]	**89.51**	**83.84**	**86.58**	85.30	**67.76**	**75.53**	**77.14**	**66.26**	**71.29**	**83.26**	**73.49**	**78.07**
YOLO-HFDM (Ours)	**87.97**	**92.64**	**90.24**	**85.32**	**80.71**	**82.95**	75.63	**77.98**	**76.79**	**83.07**	**83.53**	**83.30**

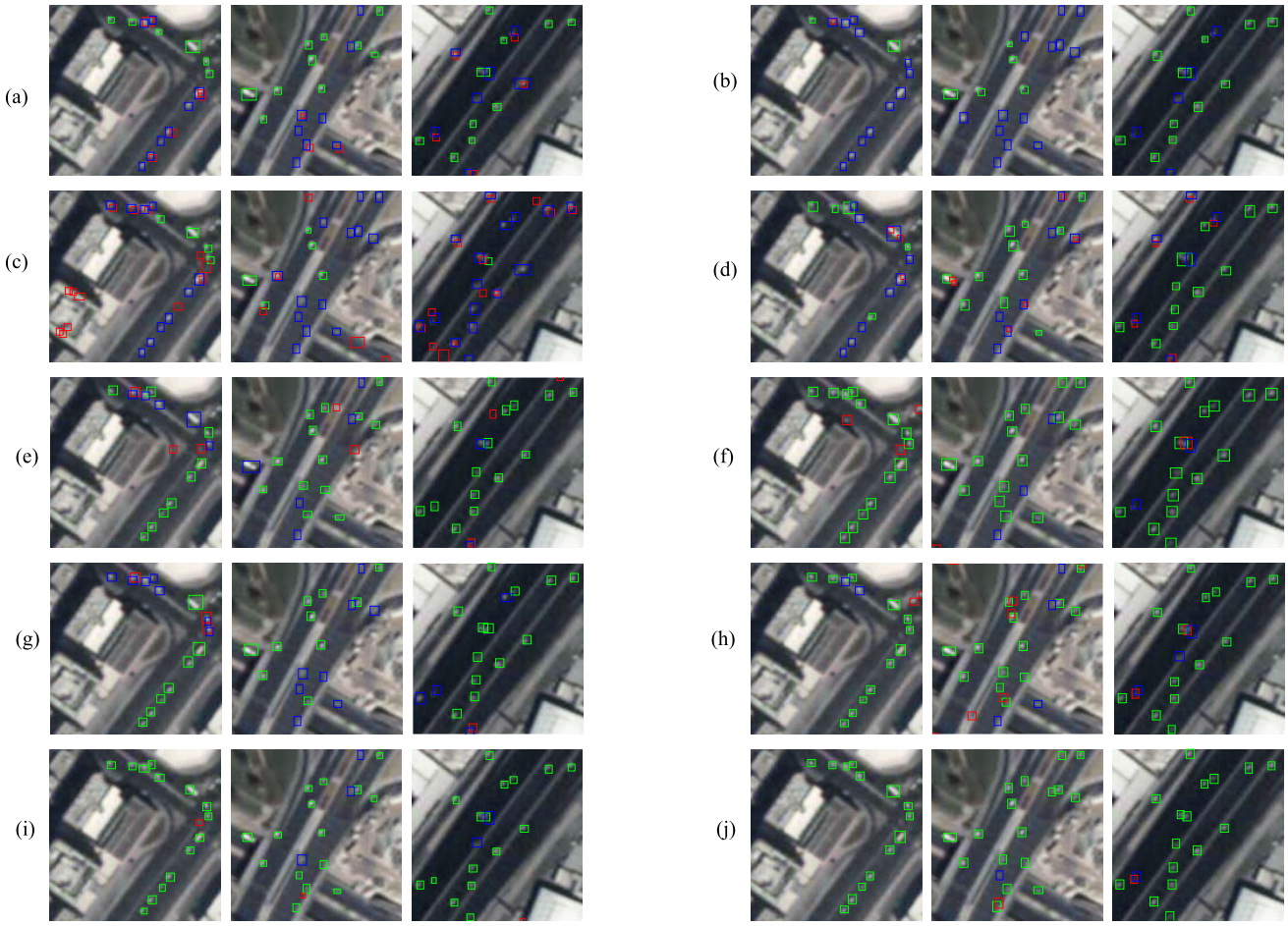


Fig. 12. Qualitative results of different comparison methods in test dataset II, each row of scenes are from three different Jilin-1 test dataset II. (a) Vibe [39]. (b) T2FGMM_UM [40]. (c) DPAdaptiveMedian [41]. (d) MOGv2 [42]. (e) CenterNet-ResNet50 [25]. (f) YOLOv5 [14]. (g) YOLOX [23]. (h) D&T [59]. (i) DSFNet [13]. (j) YOLO-HFDM (Ours). Green boxes: TP. Red boxes: FN. Blue boxes: FP.

method combines HFDM with grouped feature representation, which can effectively extract motion features of continuous frames, making them more robust to the variation of IoU.

To verify the performance of the different comparison methods on various datasets, we choose the Dubai dataset to test the performance of the model, the test results are depicted in Fig. 12 and Table III. In the three videos, Vibe [39], T2FGMM_UM [40], and DPAdaptiveMedian [41] have more miss detections since this portion of the vehicle is at an intersection, the vehicle moves slowly, and it is difficult to establish the background template accurately using the background-based differential and background modeling methods. In addition, there are more false alarms at the edges of the buildings. After parameter optimization, the precision of MOGv2 [42] is greatly improved compared with other traditional methods, but it still does not meet the

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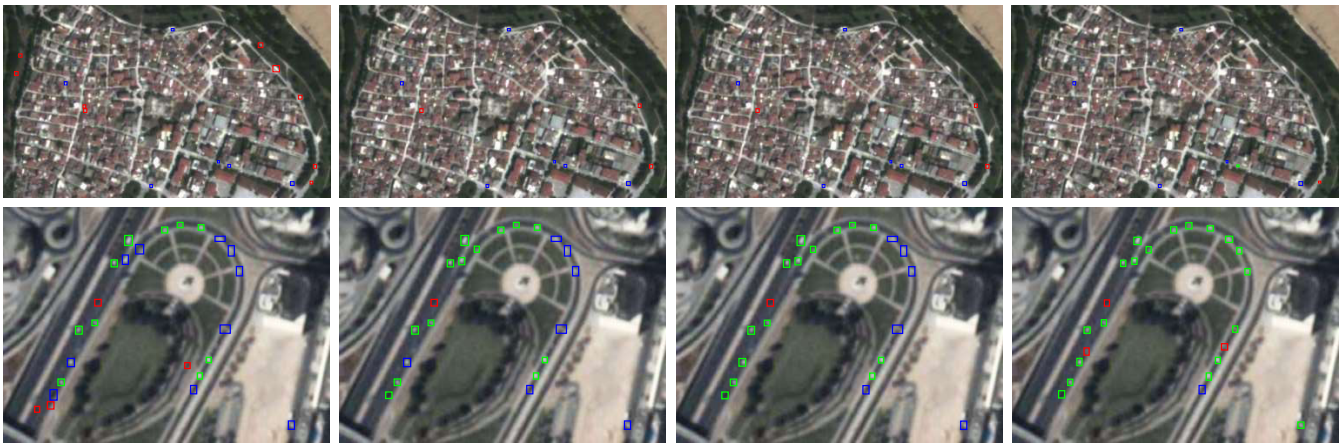


Fig. 13. Comparison with YOLOv8 and MOT. Green boxes: TP. Red boxes: FN. Blue boxes: FP.

requirements of high-precision real-time detection. D&T [59] has the phenomenon of missing detection in areas with weak feature texture structures. CenterNet [25], YOLOv5 [14], and YOLOX [23] have poor detection performance in dense vehicle areas, the vehicle location and bounding boxes cannot be precisely determined. The DSFNet [13] combined with interframe information can achieve high detection accuracy and overcome the problem of dense and difficult localization of vehicles to a certain extent, but there are also false alarms and miss detections on some weak texture feature targets. The method in this article achieves the best detection performance on the test dataset II, with the average accuracy, recall, and $F1$ score reaching 83.07%, 83.53%, and 83.30%, respectively. Experimental results show that this method can achieve state-of-the-art detection performance in complex scenarios.

Furthermore, the detection-based multiobject tracking algorithm can continuously locate vehicles. Therefore, we also compared it with this method, using YOLOv8 [24] as the object detector, ByteTrack [60], and BoT-SORT [61] for tracking. Meanwhile, to verify the HFDM module's generalization, we added it to YOLOv8. The results are shown in Table IV and Fig. 13. On the VISO and Dubai datasets, YOLOv8 has $F1$ scores of 59.97% and 73.07%. After using ByteTrack and BoT-SORT trackers, there was a slight improvement in $F1$ scores because the tracker can suppress some trajectory irrelevant detection targets based on trajectory information. YOLOv8+HFDM has shown significant improvement in both datasets, the $F1$ scores on the VISO and Dubai datasets reached 85.62% and 80.82%, respectively, demonstrating the generalization ability of HFDM on different detectors, which can improve the accuracy of motion vehicle detection in satellite videos for different detectors.

E. Ablation Study

Ablation experiments are conducted in this section to test the efficacy of three strategies, MSM data augmentation strategy, HFDM, and network structure optimization, in order to verify the efficacy of various improved strategies. The qualitative results of different comparison methods in test dataset I as shown in Fig. 14, different color bounding boxes representing the detection results of different comparison methods. The detection bounding boxes from the ablation experiments are shown in a single image. The figure shows that YOLO-HFDM can effectively improve the vehicle detection accuracy of weak color texture features in most scenarios.

1) Effectiveness of MSM: For the data augmentation strategy, based on YOLOv5, test the impact of pixel-wise, perspective, cutmix, mixup, mosaic, and MSM on model detection performance. The quantitative results are shown in Table V. The $F1$ score of the baseline is 57.08%, and the $F1$ score of the baseline with pixel-wise data augmentation is 58.16%. Based on index B, the $F1$ score decreases slightly when increasing the augmentation method of perspective, which shows that scene distortion affects the detection performance of the model to a certain extent. Cutmix, on the other hand, can better balance the accuracy and recall rates, resulting in a 0.39% increase in $F1$ score; mixup uses a mix of two images resulting in confusing scene information and, therefore, a decrease in accuracy. Mosaic uses the stitching of four images, which can better enrich the diversity of the dataset and does not cause scene distortion. The $F1$ score of the baseline with MSM is 60.18%, which shows that avoiding the generation of noninformative pixels and increasing the scale information of a single image is beneficial to improve the detection performance of the model.

2) Effectiveness of HFDM: In order to test the effectiveness of HFDM and the optimal frame number and frame interval parameters, the model performance of Baseline with and without the HFDM, different input frame numbers, and frame intervals were tested with YOLOv5 + MSM as a baseline, and the quantitative results are shown in Table VI. The model detection performance is improved by 13.59%, 15.48%, and 15.15% in precision, recall, and $F1$ score, respectively, after adding HFDM. Because in Baseline, only single-frame images are used to detect vehicles, the spatio-temporal information of satellite video still needs to be fully considered. Hence, it is difficult to effectively distinguish

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TABLE IV
QUANTITATIVE EVALUATION WITH YOLOV8 AND MOT

Methods	VISO			Dubai		
	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)
YOLOv8 [24]	59.45	60.49	59.97	72.05	74.11	73.07
YOLOv8+ByteTrack [24], [60]	65.74	63.99	64.85	79.08	73.29	76.07
YOLOv8+BoT-SORT [24], [61]	67.67	63.74	65.65	78.97	73.22	75.99
YOLOv8+HFDM	88.94	82.54	85.62	80.73	80.91	80.82

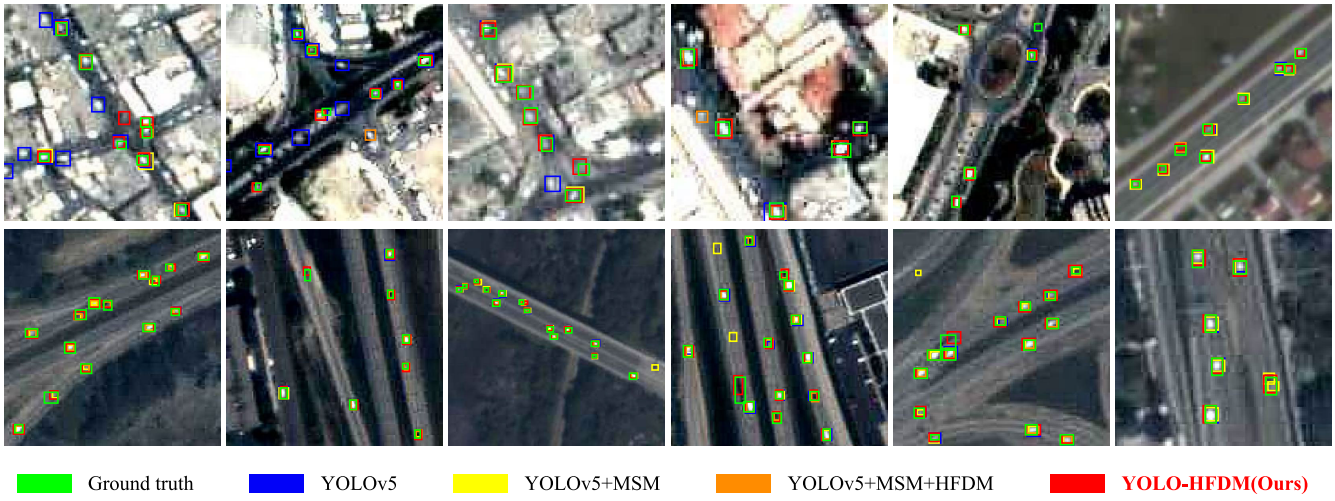


Fig. 14. Detection results of ablation experiments. MSM is the multiscene mosaic data augmentation strategy, green boxes indicate the truth bounding boxes, other different colors indicate the prediction bounding boxes of ablation experiments, and red indicates the prediction boxes of the final method in this article.

TABLE V
COMPARISON OF THE ACCURACY OF DIFFERENT DATA AUGMENTATION METHODS ON TEST DATASET I

Index	Methods	P(%)	R(%)	F1(%)
A	Baseline	74.28	46.35	57.08
B	A + Pixel-Wise	**75.17**	47.43	58.16
C	B + Perspective	74.36	47.39	57.89
D	B + Cutmix	72.15	49.27	58.55
E	B + Mixup	54.79	43.08	48.23
F	B + Mosaic	71.55	**51.53**	59.91
G	B + MSM	72.94	51.22	**60.18**

stationary vehicles from moving vehicles, and it is challenging to effectively extract vehicle information from blurred single-frame images. Therefore, in the object detection of satellite video, extracting spatio-temporal information from multiframe images can effectively improve the detection performance of the model.

In comparing models with different frame counts, the more frames used, the higher the model detection accuracy, but the detection speed also decreases. After the frame count is greater than 3 frames, the $F1$ score improvement is relatively slight, and the detection speed decreases sharply, which makes it difficult to meet the demand of real-time detection. In the comparison of frame interval, it can be seen that the model accuracy is optimal when the $\Delta t = 2$, which is because vehicles in satellite videos move relatively slowly, making it difficult to extract motion information within short intervals. In contrast, longer intervals can result in significant displacement and background offset. Therefore, in this article, $n = 3$, and $\Delta t = 2$ as the optimal parameters, considering the detection speed and accuracy.

3) Effectiveness of Network Structure: To evaluate the positive impact of grouped convolution with GA-PANet on the model architecture, we tested the effectiveness of the improvement of the method in this article using standard convolution, different backbone, and neck, respectively, and the detailed experimental results are shown in Table VII.

Table VII shows that RepVGG [26] has the same accuracy as CSPDarknet53, but the model parameters and computation are larger. CSPDarknet53-S removes a set of C3 structures and reduces the down-sampling rate of the model, and the number of model parameters decreases. However, the computation consumption increases a lot because the larger scale feature map increases the computation of the model. The accuracy performance of 0.54% increases in the $F1$ score. Using squeeze-and-excitation (SE) [57] and spatial group-wise enhance (SGE) [58] attention mechanisms in the neck can slightly improve the model accuracy. In contrast, SA [56] significantly improves accuracy. However, SA shuffles the feature map in the channel dimension, which is not conducive to the independent calculation of grouped features. GA, which is improved based on SA, has a better performance. Finally, combining grouped convolutional features allows the network model to extract the features of multiple consecutive frames more accurately and prevent the information mingling of multiple frames.

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TABLE VI
MODEL PERFORMANCE COMPARISON OF HFDM WITH DIFFERENT NUMBERS OF FRAMES AND DIFFERENT FRAME INTERVAL

HFDM	Frames (n)	Interval (Δt)	P(%)	R(%)	F1(%)	Param.	GFLOPs	FPS
Without	-	-	72.94	51.22	60.18	7.01M	15.85	44
With	2	2	86.53	66.70	75.33	7.08M	23.49	38
	3	2	91.57	80.29	85.56	7.09M	29.16	29
	4	2	92.99	79.32	85.61	7.12M	45.90	15
	5	2	93.01	79.63	85.80	7.16M	70.48	6
	3	1	91.43	77.31	83.78	7.09M	29.16	29
	3	2	91.57	80.29	85.56	7.09M	29.16	29
	3	3	90.07	80.24	84.87	7.09M	29.16	29
	3	4	89.44	77.63	83.12	7.09M	29.16	29

TABLE VII
VALIDATION OF THE NETWORK STRUCTURE

Backbone	Neck	Conv	P (%)	R (%)	F1 (%)	Param.	GFLOPs	FPS
RepVGG [26]	PANet [55]	Ordinary	92.31	79.60	85.50	10.25M	38.44	25
CSPDarknet53 [34]	PANet [55]	Ordinary	91.57	80.29	85.56	7.09M	29.16	29
CSPDarknet53-S	PANet [55]	Ordinary	93.28	79.90	86.10	3.99M	57.44	20
CSPDarknet53-S	PANet [55]	Group	93.70	80.50	86.60	3.13M	51.89	23
CSPDarknet53-S	SE-PANet [57]	Group	93.69	80.92	86.84	3.17M	52.84	21
CSPDarknet53-S	SGE-PANet [58]	Group	**93.90**	81.20	87.10	3.15M	51.91	22
CSPDarknet53-S	SA-PANet [56]	Group	93.40	82.20	87.44	3.14M	51.90	22
CSPDarknet53-S	GA-PANet	Group	93.35	**82.83**	**87.78**	3.18M	52.86	21

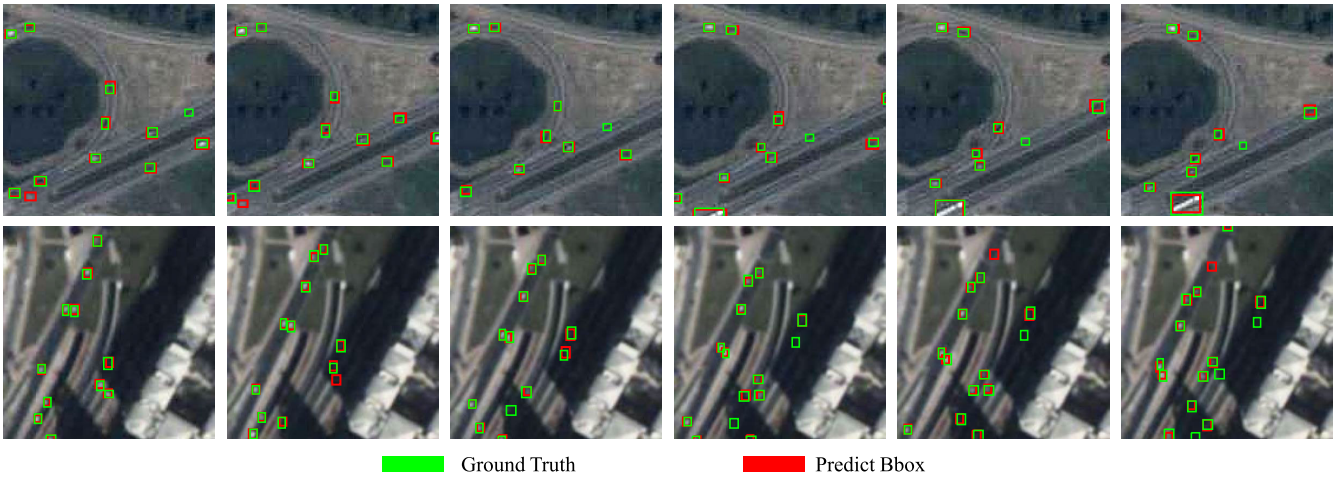


Fig. 15. Qualitative results of YOLO-HFDM on weak color texture images.

In order to further verify the ability of the method proposed in this article to resist weak color texture features, satellite video data with weak color texture was selected and visualized at specific intervals; the qualitative results are shown in Fig. 15. The figure shows that in the first region, most of the vehicles on the road are relatively blurry, making it difficult to distinguish between vehicles and the road background. YOLO-HFDM still achieves high detection accuracy and performs well in boundary localization; in the second region, building shadow occlusion has a problem, making it difficult to identify vehicles in the shadow solely based on a single frame image. However, YOLO-HFDM can improve the target localization performance with the assistance of historical information. Therefore, this method can address the problem of weak color texture features.

V. CONCLUSION

In this article, we propose a satellite video moving vehicle detection algorithm of YOLO-HFDM, which can address the problem of small moving targets and weak color texture features in satellite videos. In the data preprocessing stage, the MSM data augmentation strategy is used to improve the model

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performance in complex scenes. Then, the spatio-temporal feature information of consecutive frames is obtained using the HFDM, which can improve the expression ability of motion features, thereby addressing the problem of fewer texture features in the target structure of single-frame images. In the detection network, by using the group convolution operation, we avoid the intermixing of continuous frame information; by reducing the model down-sampling multiplicity, the use of large-scale feature maps is more favorable to the prediction of small targets; and by using GA-PANet to improve the group feature expression of multiple frames. It is shown that the method in this article can substantially improve the model recall and achieve state-of-the-art detection performance with a detection speed of 21 FPS. In comparison to the state-of-the-art method, the method presented in this article achieves the highest detection accuracy in complex scenes and satisfies the demand for real-time detection. For the information processing of continuous frames, however, further research is needed to explore in depth the application of different differential feature processing methods in satellite video moving vehicle detection in the future.

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